

Balancing Chemical Equations

Answer key with worked solutions

A. Understanding

1. Why balance?

Atoms are rearranged in a reaction, never created or destroyed (**conservation of mass**), so each element must have the same number of atoms on both sides.

2. Coefficient vs subscript.

The 2 in $2\text{H}_2\text{O}$ is a **coefficient**: it means two water molecules. The 2 in H_2O is a **subscript**: two H atoms in each molecule, part of what makes it water. Only coefficients may be changed; changing a subscript would change the substance (e.g. H_2O_2 is hydrogen peroxide).

B. Counting atoms

3. Count atoms.

(a) 4 N, 12 H (b) 3 Mg, 6 O, 6 H (c) 4 Fe, 6 S, 24 O

C. Balancing

4. Balance.

(a) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ (b) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (c) $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ (d)
 $2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$

5. Balance.

(a) $4\text{Al} + 3\text{O}_2 \rightarrow 2\text{Al}_2\text{O}_3$ (b) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (c) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$

6. Combustion.

(a) $\text{C}_2\text{H}_4 + 3 \text{O}_2 \longrightarrow 2 \text{CO}_2 + 2 \text{H}_2\text{O}$ (b) $\text{C}_2\text{H}_5\text{OH} + 3 \text{O}_2 \longrightarrow 2 \text{CO}_2 + 3 \text{H}_2\text{O}$. Remember the O already in ethanol: right side $4 + 3 = 7$ O, so $7 - 1 = 6$ O from O_2 . (c) $2 \text{C}_4\text{H}_{10} + 13 \text{O}_2 \longrightarrow 8 \text{CO}_2 + 10 \text{H}_2\text{O}$. With 1 C_4H_{10} you would need $6\frac{1}{2} \text{O}_2$, so double everything.

7. Polyatomic ions.

(a) $2 \text{AgNO}_3 + \text{CaCl}_2 \longrightarrow 2 \text{AgCl} + \text{Ca}(\text{NO}_3)_2$ (b) $\text{H}_2\text{SO}_4 + 2 \text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + 2 \text{H}_2\text{O}$ (c) $\text{Al}_2(\text{SO}_4)_3 + 3 \text{Ca}(\text{OH})_2 \longrightarrow 2 \text{Al}(\text{OH})_3 + 3 \text{CaSO}_4$

D. From words to equations

8. Sodium and water.

$2 \text{Na} + 2 \text{H}_2\text{O} \longrightarrow 2 \text{NaOH} + \text{H}_2$. Check: Na $2 = 2$, H $4 = 4$, O $2 = 2$.

E. Challenge

9. Check and balance.

(a) **Not balanced:** H and Cl are 1 on the left but 2 on the right. Balanced: $\text{CaCO}_3 + 2 \text{HCl} \longrightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ (b) $6 \text{CO}_2 + 6 \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2$. Check: C $6 = 6$, H $12 = 12$, O $18 = 18$.